A-A-53954
31 August 1989

COMMERCIAL ITEM DESCRIPTION

BUR, DENTURE TRIMMING (Straight Handpiece, Tungsten Carbide, Staggered-Tooth)

The General Services Administration has authorized the use of this Commercial Item Description.

Salient Characteristics:

This commercial item description covers staggered tooth tungsten carbide denture trimming burs for use in a straight handpiece, in the following shapes, as specified.

Torpedo
Bud
Cylinder
Pear

Material.

One-piece bur. The torpedo shaped bur shall be fabricated entirely of sintered tungsten carbide/cobalt. The fine grit tungsten carbide shall be submitted to hot isostatic processing.

Two-piece bur. The head of the two-piece bur (other than torpedo) shall be fabricated from sintered tungsten carbide/cobalt. The fine grit tungsten carbide shall be submitted to hot isostatic processing. The neck, shank and base shall be fabricated from hardened Class 4, Type 420 C martensitic corrosion-resisting steel as specified in ASTM F899, for composition only, or chromium or nickel plated AISI-S5 alloy steel as specified in ASTM A681, for composition only.

Hardness. The entire torpedo shaped bur and the bur heads for the remaining burs shall have a Vickers hardness of 1600 as specified herein for a 3 kgf. The shanks of the remaining burs shall have a Rockwell hardness of C52-C54 if fabricated from type 420 corrosion-resisting steel. If fabricated from AISI-S5 alloy steel, shall have a Rockwell hardness of C58 minimum, as specified herein.

AMSC N/A FSC 6520

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Style, design and dimensions. Style, design and dimensions shall be in accordance with Defense Personnel Support Center (DPSC) Drawing 22420. The burs shall be used in a straight handpiece for trimming, carving, and finishing chrome cobalt, acrylic, and dental stone. The two-piece burs shall be permanently welded or brazed.

Blades. The blades shall have a fine staggered toothing, cutting to the right.

Torpedo. There shall be twenty-two (22) blades with right-hand twist and eighteen (18) blades with left-hand twist.

Bud and cylinder. There shall be twenty (20) blades in five (5) groups with right-hand twist and sixteen (16) blades with left-hand twist.

Pear. There shall be sixteen (16) blades in four (4) groups with right-hand twist and fourteen (14) blades with left-hand twist.

The blades of the bur head shall be shaped and sharpened to cut at the tip as well as the sides of the bur head, and to provide automatic chip clearance.

Runout. The total indicated runout of the burs shall not exceed 0.002 inch, when tested as specified herein.

Cutting test. The time required to cut the eighteenth one-half inch shall be not more than 2.25 times the time that is required to cut the second one-half inch, when tested as specified herein. The burs shall show no breaking, bending, or loosening of the head as a result of these tests.

Corrosion resistance. The burs shall be corrosion-resistant when using cleansing and disinfectant liquids for rotary dental instruments, under the precondition that the manufacturer's instructions with regard to the application of those liquids are observed.

Sterilization. The burs shall withstand 1 (one) steam autoclave as specified herein.

Finish. The bur heads shall have a finely honed finish with sharpened blades. The joint and shank of the burs shall have a smooth and bright finish.

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Identification marking. Marking of the burs with the name or registered trademark of the manufacturer is optional.

Workmanship. The burs shall be free from pitting, coarse grind marks, incomplete welding or any other defects which may detract from their appearance or impair their serviceability.

Unit. Each (EA). One bur, of the shapes specified, constitutes one unit.

Unit package. One bur shall be a single package: a plastic base with a clear plastic cover.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Sampling for inspection. Sampling for inspection shall be conducted in accordance with MIL-STD-105. Where feasible, the same sample shall be used for the determination of two or more inspection characteristics. The unit of product for inspection purposes shall be one bur of each shape. Unless otherwise specified, sampling for examination shall be conducted at inspection level II, except for dimensions which shall be conducted at inspection level S-1, and sampling for tests (when applicable) shall be conducted at inspection level S-1.

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Examination. Examination shall include, but shall not be limited to the following:

Defects

Bur head shape not as specified.
Number of blades not as specified.
Dimensions of the burs not within tolerance limits.
Welded or brazed joint of bur does not give the appearance of an integral head unit.
Finish on bur not as specified.

Dimensions. Examination of the dimensions shall be made with a toolmaker's microscope or other instrument of equal accuracy. Manufacturer's gages may be used if properly calibrated checking standards are used in conjunction with them.

Microscopic examination. Examination of the bur defects not readily perceptible to the naked eye shall be made using a microscope at 10 power (10x) magnification.

Test procedures.

Chemical composition. In the absence of certificates of quality for the raw steel used in the fabrication of the bur mandrels, chemical analysis of the mandrels shall be determined in accordance with Method 111 or 112 of FED-STD-151. In case of a dispute, method 111 shall be used for referee analysis.

Hardness. Hardness for the entire torpedo bur and the remaining bur heads shall be in accordance with ASTM E92. The shanks of the remaining burs shall be in accordance with ASTM E18.

Autoclaving. The burs shall be steam autoclaved at a temperature of 270°F (132 degrees Celsius) at 27 psi for 15-20 minutes.

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Runout. The mandrel of the bur shall be placed and held in a "V" block or precision chuck, and rotated slowly by any suitable means. A dial indicator sensitive to 0.0002 inch and equipped with a knife-edge foot, or a shadow-graph, or other suitable equipment of equal accuracy, shall be employed. If a dial indicator is used, the indicator shall be mounted with the shaft perpendicular to the central axis of the mandrel of the bur, and so positioned that the foot contacts the mandrel of the bur at a point 1/8 ± 1/32 inch beyond the base of the head (where the mandrel contains no cement). The maximum and minimum readings obtained during one complete 360-degree rotation of the bur shall be measured and recorded to the nearest 0.0005 inch. The difference between these two readings shall be taken as the total indicated runout.

Cutting test. Eighteen half-inch cuts shall be made in a 0.020 inch thick brass strip (Brinell hardness 74-81) at 12,000 revolutions per minute and a 2 pound load.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producer's own drawings, specifications, standards and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch/pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerance specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

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Regulatory requirements.

Federal Food, Drug, and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580, as amended, and as implemented by Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

NOTE: The following National Stock Numbers are covered by this document:

NATIONAL STOCK NUMBERS	ITEM IDENTIFICATION
6520-01-283-1072	BUR, DENTURE TRIMMING, Straight Handpiece, Tungsten Carbide, Pear
6520-01-283-1073	BUR, DENTURE TRIMMING, Straight Handpiece, Tungsten Carbide, Cylinder, Round

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6520-01-283-1074	BUR, DENTURE TRIMMING, Straight Handpiece, Tungsten Carbide, Torpedo
6520-01-283-7967	BUR, DENTURE TRIMMING, Straight Handpiece, Tungsten Carbide, Bud

Ordering data. Intermediate/exterior package quantities and labeling and marking must be as specified in the contract and/or order.

MILITARY INTERESTS:

Army-MD
Navy-MS
Air Force-03

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
PHS

Project No. 6520-2230
Location- NCR Disk 1

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